

3D picking for Universal Robots. A URCap whose 3D Pick program node uses a depth camera on the wrist to find every part of the size you enter, number them in the pick order you choose, and move the tool to the grip on each one. Your program opens and closes the gripper.

Planning figures, not guarantees. The numbers below are deliberately conservative estimates derived from the software's limits, its programmed speeds and the camera's noise. This release is a beta and the figures have not been measured on a production cell. **Test with your parts, gripper and layout before you commit to them.**

PICKING PERFORMANCE (by camera-to-part distance; verify on your cell)

	0.30 to 0.60 m working zone	1.0 to 1.5 m far limit	
Parts found	99 % ^S	4 of 4 ^M	Right-size parts in view. Wrong ones picked: 0.
Size resolved to	±5 mm	±17 mm	It checks a size; it does not measure one.
Sizes told apart	≥ 10 mm ^E	≥ 35 mm ^E	Closer sizes read as the same part.
Smallest side	20 mm ^E	45 mm ^E	Edges blur by ~1 % of the distance.
Lowest part	5 mm ^E	15 mm ^M	Above the surface; rough ones raise it.
Part centre	±3 mm ^S	±10 mm ^E	95 %, plus calibration: 9 mm per 1°.
Part heading	±2.5° ^S	±8° ^E	95 %. None for a square top.
Gap to neighbours	≥ 25 mm	≥ 40 mm ^E	20 mm finger room a side.
Time to find	0.9 s ^M	1.1 s ^M	Per picture, Raspberry Pi 5.
Distance limits	0.28 to 1.6 m	No depth nearer; nothing farther is picked.	
Grip position	±10 mm ^E	Plan the jaws for it. Mostly calibration.	
Part pose	Any face down	Boxes on any face; cylinders on end.	
Cycle, node only	15 s per part ^E	To the grip. ~2 parts/min with a short place.	

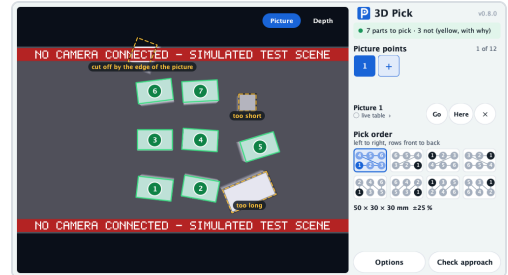
^M Measured with the camera computer and a D435, one setup. ^S Simulated: 720 parts through a model of the D435's noise. ^E Estimate, not a measured or guaranteed value.

NOT FOR (LIMITS OF THE DEPTH CAMERA)

- Measuring parts or quality checks
- Defect, surface or print inspection
- Telling apart parts of nearly the same size
- Thin or flat parts: sheets, washers, labels
- Transparent, mirror-like or very dark parts
- Bin picking: piled or leaning parts
- Shapes other than boxes and upright cylinders
- Direct sunlight on the work area
- Driving the gripper or the place move

CAPABILITIES

- Finds parts by volume, from depth alone:** boxes on any face, cylinders on end; the floor is fitted round each part.
- Any position, any rotation.** Fingers across the short side, or the long one.
- Eight pick orders:** by rows or by columns, from any corner.
- Checks before it moves:** reach by the arm's kinematics, finger room.
- Says why not:** near misses drawn in yellow with the reason.
- Queues what it saw:** no trip back to the picture point.
- One program node:** sets rs_pick_found, rs_pick_loc. No internet.



3D Pick on a PolyScope 5 pendant (simulated scene).

COMPATIBILITY

PolyScope 5	5.4 to 5.26 (e-Series)
PolyScope X	10.8 to 10.14
Robots	UR3e, UR5e, UR7e, UR10e, UR12e, UR16e, UR20
Gripper	Any your program can open and close. The node does not drive it.

WHAT A CELL NEEDS

Camera	RealSense D435 on a printed bracket between the tool flange and the gripper
Computer	Small Linux computer beside the robot (Raspberry Pi class), on the robot's network
Network	TCP 7621 and 7622, robot to camera computer
Setup	One camera-to-flange calibration; optional pick areas taught with three fingertip touches
URCap	Free, open source (MIT). Kit from Advin.

Download and details: perceptronics.advin.io · **Kit and questions:** nick@advin.io · **Source:** github.com/JimothyJohn/perceptronics

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